TECH BIZ SECURITY · SOC ANALYST INTERNSHIP · BATCH B-503

WEEK 1 · TASK 2

Networking Basics for Defenders

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01 Objective

A SOC analyst reads network traffic every day, so I need to know what normal looks like before I can spot what is wrong. This week I studied networking from a defender's point of view: how devices are identified (IP and MAC addresses), how data travels through the seven OSI layers, which ports and protocols matter most in security work, and how DNS turns a name into an address. I finish by contrasting normal and suspicious traffic, which is the judgement a Tier 1 analyst has to make all day.

02 Proof of Completion

Room / resource	Duration	Tasks	Status
TryHackMe: What is Networking?	30 min	5	Completed, room progress 100%
TryHackMe: DNS in Detail	45 min	5	Completed, room progress 100%
OSI Model (alternative source)	-	7 layers	Studied via SentinelOne

> **Note on the OSI Model room**
> The TryHackMe OSI Model room is locked behind a subscription. As the task instructions allow, I did not use it. I used the free SentinelOne article "OSI Model" instead, and I also mapped each layer to protocols and attacks in section 04.

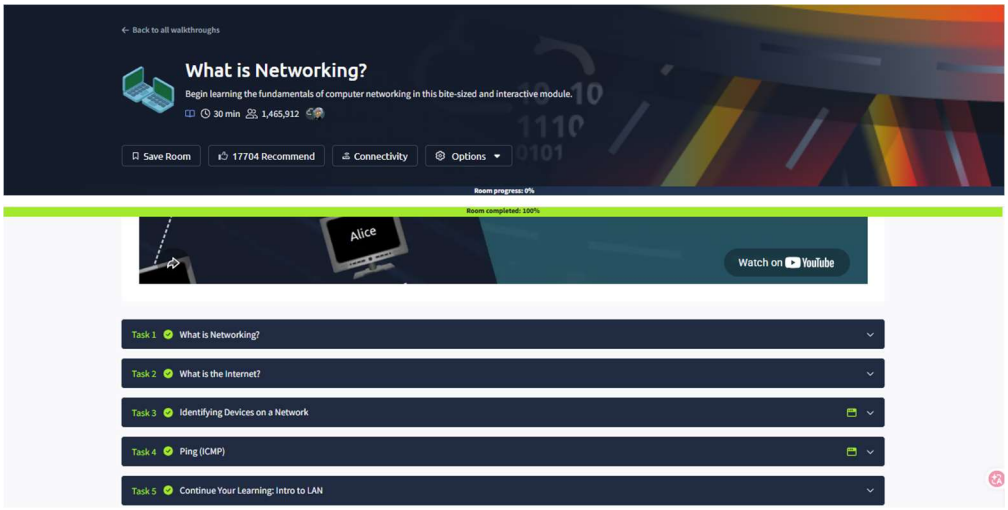


Figure 1. What is Networking?: all 5 tasks completed (100%).

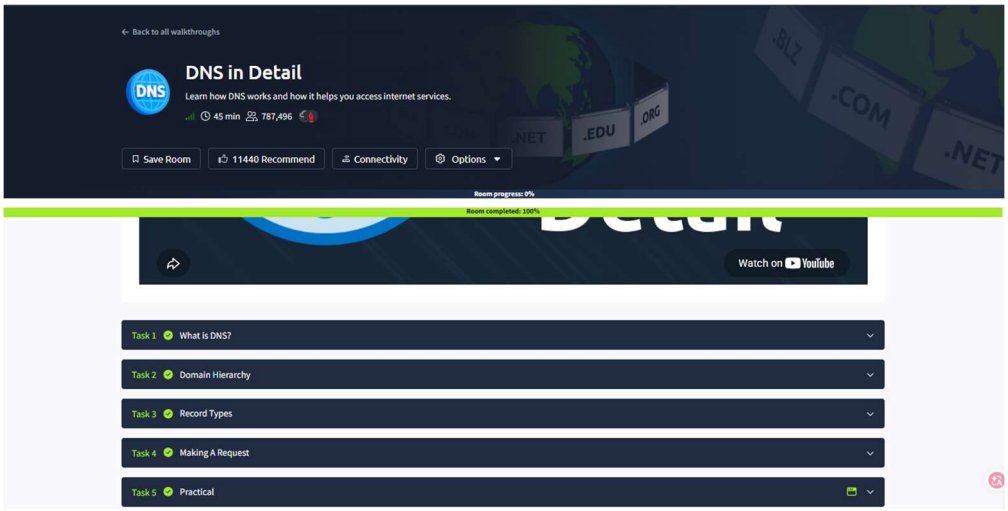


Figure 2. DNS in Detail: all 5 tasks completed (100%).

03 Networking Foundations

The first room gave me the starting point: a network is simply things connected, and the internet is a very large network of networks. Every device needs two kinds of identity to take part.

Concept	What it is	Why a defender cares
IPv4 address	32-bit logical address, written as four numbers (for example 192.168.1.10). About 4.3 billion exist.	The first thing I check when tracing a source or destination in a log. Private ranges (10.x, 172.16-31.x, 192.168.x) stay inside the LAN.
IPv6 address	128-bit address written in hexadecimal. It was created because IPv4 addresses are running out.	Often forgotten by monitoring rules, so IPv6 traffic can slip past defences tuned only for IPv4.
MAC address	48-bit hardware identifier of a network card, used inside a local network.	Useful for identifying a physical device, but easy to spoof.
Ping (ICMP)	A test that sends an echo request and waits for an echo reply to check that a host is reachable.	Attackers use ping sweeps to find live hosts, and ICMP can be abused to hide data.
LAN	A local network that connects devices in one place, such as an office.	Traffic inside the LAN is where lateral movement happens.

> **IP vs MAC in one line**
> The IP address says where a device is on the network. The MAC address says which physical device it is on the local link.

04 The OSI Model (7 Layers)

The OSI model splits network communication into seven layers. Data moves down the layers on the sender's side, gaining a header at each step (encapsulation), and moves back up on the receiver's side. A handy way to remember the order from layer 7 down to layer 1 is: All People Seem To Need Data Processing.

	Layer	What it does	Data unit	Common protocols	Typical attack
7	Application	Where users and programs reach network services; the layer people interact with.	Data	HTTP, DNS, SMTP, FTP	Phishing, SQL injection, DNS tunnelling
6	Presentation	Defines how data is encoded, encrypted and compressed for sender and receiver.	Data	TLS/SSL	SSL stripping
5	Session	Opens, coordinates, maintains and closes conversations between applications.	Data	NetBIOS, RPC	Session hijacking
4	Transport	Splits data into segments and reassembles them at the other end.	Segment	TCP, UDP	SYN flood, port scanning
3	Network	Breaks segments into packets and routes them along the best path.	Packet	IP, ICMP	IP spoofing, ping sweeps
2	Data Link	Links two directly connected devices and sends frames between them.	Frame	Ethernet, ARP	ARP spoofing, MAC flooding
1	Physical	Carries raw bits over electrical, radio or optical media.	Bits	Cables, Wi-Fi radio	Cable tapping, physical theft

> **How I will use this as an analyst**
> When an alert comes in, I ask which layer it lives on. A port scan is a layer 4 problem, an ARP spoof is layer 2, and a phishing link is layer 7. Naming the layer tells me which logs to open next.

TCP/IP in practice: real networks use the simpler four-layer TCP/IP model, but the OSI vocabulary is the one analysts use when they discuss problems.

05 Ports and Protocols

A port is a numbered door on a host that tells the system which service should receive the traffic. The six ports the task asked for are first, followed by four more that come up constantly in SOC work.

Port	Service	Transport	Purpose	What I watch for as a defender
22	SSH	TCP	Encrypted remote login and file transfer	Brute-force attempts, logins from unusual countries
80	HTTP	TCP	Unencrypted web traffic	Malware downloads, plain-text credentials, web attacks
443	HTTPS	TCP	Encrypted web traffic	Command-and-control traffic hiding inside normal-looking encryption

Port	Service	Transport	Purpose	What I watch for as a defender
53	DNS	UDP / TCP	Translates domain names to IP addresses	Tunnelling, very long or random queries, newly seen domains
445	SMB	TCP	Windows file and printer sharing	Lateral movement and ransomware spreading between hosts
3389	RDP	TCP	Windows remote desktop	Brute force, and RDP exposed to the internet
21	FTP	TCP	File transfer, unencrypted	Credentials and files sent in clear text
23	Telnet	TCP	Remote login, unencrypted	Should not be used; any use is a finding
25	SMTP	TCP	Sending email between servers	Spam, phishing and data leaving by mail
5985	WinRM	TCP	Windows remote management over HTTP	Remote command execution used for lateral movement

TCP vs UDP

	TCP	UDP
Connection	Connection-oriented; starts with a three-way handshake (SYN, SYN-ACK, ACK).	Connectionless; sends data with no handshake.
Reliability	Confirms delivery and resends lost data.	No delivery guarantee.
Speed	Slower because of the extra checks.	Faster and lighter.
Used for	Web, email, SSH, SMB.	DNS queries, streaming, voice.

06 How DNS Works

DNS is the internet's phone book: it turns a name that people remember into the IP address that computers need. Because almost every connection starts with a DNS lookup, DNS logs are one of the best places to see what a machine is trying to reach.

6.1 Domain hierarchy

Take the address academy.techbiz-sec.com. Read from right to left, it breaks down into three parts:

Part	Example	Meaning
Top-level domain (TLD)	.com	The last label. It is either a gTLD (generic, such as .com or .org) or a ccTLD (country code, such as .pk or .fr).
Second-level domain	techbiz-sec	The name registered by the owner.
Subdomain	academy	A section the owner creates under the main domain.

6.2 Resolution flow

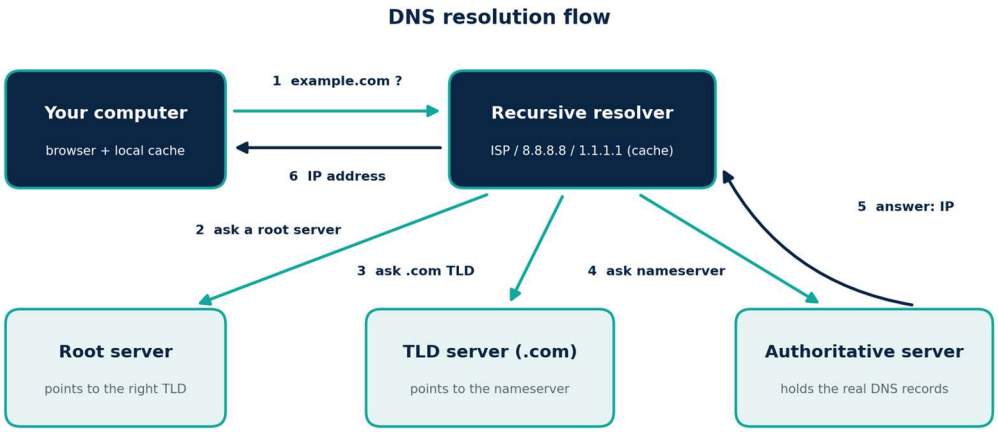


Figure 3. The path of a DNS request from my computer to the answer.

- **Step 1:** my computer checks its local cache. If the name is not there, it asks the recursive resolver, usually provided by my ISP.
- **Step 2:** the resolver checks its own cache. If it has no answer, it starts at a root server, which points to the correct TLD server.
- **Step 3:** the TLD server holds the records that say which nameserver is responsible for the domain.
- **Step 4:** the authoritative server, the one that stores the domain's records, returns the real answer.
- **Steps 5-6:** the resolver caches the result and gives the IP address to my computer, which then connects.

6.3 Record types

Record	What it returns	Security relevance
A	An IPv4 address (for example 104.26.10.229).	The most common lookup; check the returned IP against threat intelligence.
AAAA	An IPv6 address (for example 2606:4700:20::681a:be5).	Same as A, but easy to overlook in monitoring.
CNAME	Another domain name (an alias).	Attackers can chain aliases to hide the final destination.
MX	The mail servers for the domain.	Needed to understand where phishing mail may come from or go to.
TXT	Free-form text data.	Holds SPF, DKIM and DMARC email protections, but is also abused to carry hidden data.
NS	The authoritative nameservers for the domain.	Changes here can mean a domain was hijacked.
PTR	A hostname for an IP address (reverse lookup).	Helps put a name to an IP during an investigation.

07 Normal vs Suspicious Traffic

The task asked for one example. I chose DNS, because it is allowed through almost every firewall and so attackers like to hide inside it.

Indicator	Normal	Suspicious
Query pattern	A few lookups for known sites such as office or update services.	Hundreds of queries per minute to one unfamiliar domain.
Domain names	Readable names such as mail.company.com.	Very long random-looking subdomains such as x9k2qa7f.badsite.xyz.
Record types	Mostly A and AAAA.	Unusual volume of TXT queries.
Timing	During working hours, tied to a user action.	Steady beaconing at night with nobody at the keyboard.
Domain age	Long-established and reputable.	Newly registered, or flagged by threat intelligence.

How I would triage it

- **Identify:** which host and which process made the queries?
- **Enrich:** check the domain and resolved IP in VirusTotal or another threat-intelligence source.
- **Scope:** search for the same domain on other hosts.
- **Decide:** close as a false positive with a note, or escalate to Tier 2 with the evidence attached.

A second example: one workstation connecting on port 445 to many other workstations in a short time is not how people normally use file sharing. It looks like lateral movement or a worm.

08 Reflection and Key Takeaways

- Networking is the map I will read every alert against. If I cannot say which layer, port and protocol an alert is about, I cannot triage it.
- The same service can be normal or malicious depending on volume, timing and destination, so context matters as much as the protocol.
- DNS logs show where a machine is trying to go, even when the connection itself is encrypted.
- Learning the attacks that belong to each OSI layer connects this task to the threat concepts from Task 1.

Next step: practise reading real packet captures and DNS logs so these patterns become second nature.

09 References

- TryHackMe: What is Networking? (free room)
- TryHackMe: DNS in Detail (free room)
- SentinelOne: OSI Model (free alternative to the locked TryHackMe room)